CS & IT ENGINEERING



Algorithms

Analysis of an Algorithms
Discussion Notes

DPP – 02



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[MSQ] → multiple can be correct

#Q. Which of the following notation is/are transitive but not reflexive

Ans :- (c, d)

A. Big oh (O) ✗

B. Big omega (Ω) ✗

C. Small oh (o) ✓

D. Small omega (ω) ✓

	O	Ω	Θ	o	ω
Reflexive	✓	✓	✓	✗	✗
Transitive	✓	✓	✓	✓	✓

[MSQ]

#Q. If $f(n) = \sum_{i=1}^{n} i^3$

Then which of the following choices is/are true for f(n)?

A $\theta(n^4)$ ✓

B $\Omega(n^4)$ ✓

C [illegible] ✓

D $\Omega(n^3)$ ✓

Tight LB

$\Omega(n^4)$

$O(n^5)$

$\theta(n^4) \rightarrow O(n^4) \rightarrow$ Tight LB

Ω

Ans:- (A, B, C, D)

$O(n^5, n^6, n^7)$ Loose LB

$n^3, n^2, n, \sqrt{n}$ Loose LB

$$f(n) = \sum_{i=1}^{n} i^3$$

$$= 1^3 + 2^3 + 3^3 + \cdots + n^3$$

$$= \left(\frac{n(n+1)}{2}\right)^2$$

$$= \frac{n^2(n+1)^2}{4} = \frac{n^2(n^2+2n+1)}{4}$$

$$f(n) = \frac{n^4 + 2n^3 + n^2}{4}$$ ✓

$O(n^4)$ ✓ $\Rightarrow$ $\Theta(n^4)$ ✓

$\Omega(n^4)$ ✓

[MCQ]

#Q. Consider the following program:

```
main ( )
{
    P = n!
    for (i = 1; i ≤ n ; ++i)
  for (j = 1 ; j ≤ P ; 2*j )
        C = C + 1;
}
```

What is the time complexity of above code?

P ⇒ n^n

O(n)

n! ≈ n × (n − 1) × ... 1 ≈ n^n

A $O(n^2)$

B $O(n^2 \log n)$

C $O(n \log n)$

D $O(n)$

Ans : B

```
for( j=1; j<= p ; j=2*j )
```

$1 \to 2^1 \to 2^2 \to 2^3 \cdots 2^k$ (↓ p)

$2^k = p$

$k = (\log_2 p)$

Total TC ⇒ $O(n * \log_2 p)$

$\log_2 p = \log_2 (n^n)$

$= n \log n$

TC ⇒ $O(n \times n \log n) = O(n^2 \log n)$

[MCQ]

#Q. Consider the following code:

```
main ( )
{
        i = 1; j = 1
        while (j ≤ n )
        {
              ++ i;
              j  = j + i;
            }
```

What is the time complexity of above code?

Ans :- B

A $\theta(n)$ ✗

B $\theta(\sqrt{n})$ ✓

C $\theta(\log)$ ✗

D $\theta(n\log(\log n))$ ✗

initially → 1st iter → 2nd iter → 3rd iter

$i = 1$ $\quad i = 2$ $\quad i = 3$ $\quad i = 4$

$j = 1$ $\quad j = 1+2$ $\quad j = 1+2+3$ $\quad j = 1+2+3+4$

Kth iter:

$i = 1, k+1$

$j \rightarrow 1+2+3+4 \ldots \quad (k+1)$

$\Downarrow n$

$$\sum_{i=1}^{k+1} i = n$$

$$\frac{(k+1)(k+1+1)}{2} = n$$

$$k^2 \approx O(n)$$

$$\boxed{k = O(\sqrt{n})}$$

[MCQ]

#Q. Consider the following code:

```
Algorithm T(n)
{
        if (n = 1) return;
        else
{
        T(n/2);
        }
    }
```

What is the space complexity of above code?

Ans :- A

A $\theta(\log n)$

B $\theta(n)$ ✗

C $\theta(n \log (\log n)$ ✗

D $\theta(\sqrt{n})$ ✗

Space Complexity → Auxiliary Space
(Additional) Space.

Recursion → Recursion Stack.

↓

Space Complexity

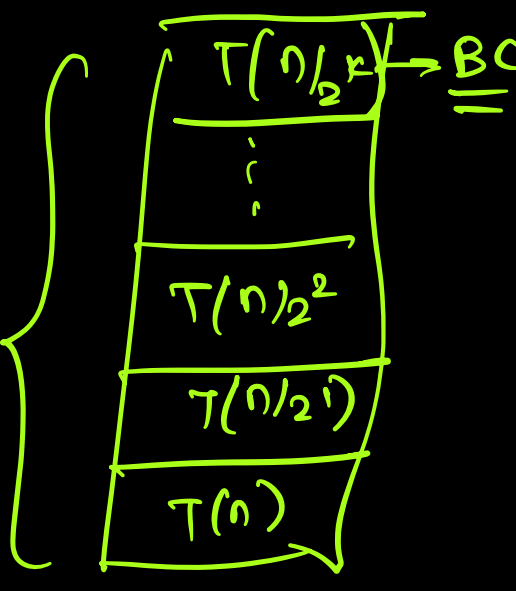
TC ⇒ $T(n) = T(n/2) + a$

$= T(n/2^2) + 2a$

⋮

$T(n) = \log_2(n)$

Recursion stack



For Base Condition
(To stop Recursion)

$\frac{n}{2^k} = 1$

$2^k = n$

$k = \log_2 n$

Size of Recursion Stack = 'k'

$O(k)$

[MSQ]

#Q. $f(n)=2^{n^2}$, $g(n)=n!$ $h(n)=2^{\log n^2}$

Asymptotic Comparison.

Which of the following is/are correct?

A. f(n) = Ω (g(n)) ✓

B. h(n) = Ω (g(n)) ✗

C. h(n) = O (g(n)) ✓

D. g(n) = Ω (f(n)) ✗

Conclusion: $f > g > h$

(A) $f = \Omega(g)$ ✓ $f \geq g$

(B) $h = \Omega(g)$ $h \geq g$ ✗

(C) $h = O(g)$ $h \leq g$ ✓

(D) $g = \Omega(f)$ $g \geq f$ ✗

Ans: A, C

$f(n) = 2^{n^2}$

$g(n) = n! \approx n^n$

$h(n) = 2^{\log_2 n^2}$

$= n^2$

$f = 2^{n^2} \quad > \quad g = n^n$

$\log_2(2^{n^2}) \qquad \log_2(n^n)$

$n^2 \qquad\qquad \not\asymp \log n$

$n \quad > \quad \log n$

$g = n^n \qquad h = 2^{\log_2 n^2} = n^2$

$n^n \quad > \quad n^2 \qquad\qquad g > h$

Conclusion $\boxed{f > g > h}$

[NAT]

#Q. Consider the following notations:

1. $\sqrt{\log n} = O(\log \log n)$ → False.
2. $\log n = \Omega\left(\frac{1}{n}\right)$ → true
3. $n^2 = \theta\left(2^{2\log n}\right)$ → True
4. $(0.061)^n = \theta(1.02)^n$ → False

How many notations is/are correct? _____.

Asymptotic Comparison.

$(0.061)^n \leq c\,(1.02)^n$

$\frac{1}{2} \to \left(\frac{1}{2}\right)^2 \to \frac{1}{4}$

① $\sqrt{\log n} = O(\log(\log n))$ ✗

Falsl

$(\log n)^{1/2} \leq C * \log(\log n)$

$(\log n)^{1/2} > \log(\log n)$

Take log both sides

$\frac{1}{2}\log(\log n) \quad \log(\log(\log n))$

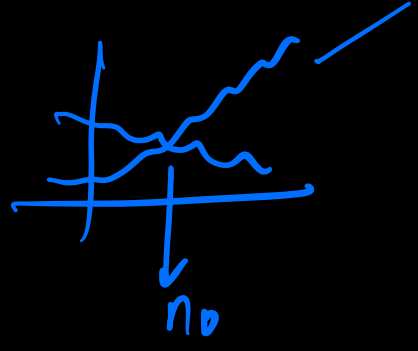
$\log(\log n) = a$

$\frac{a}{2} > \log(n)$

② $\log n = \Omega(1/n)$

$\log n \geq \frac{C*1}{n}$ $\quad 1/n \rightarrow$ Decr

incr. $\quad \hookrightarrow$ True



③

(3) $n^2 = \Theta(2^{2\log n})$

$n^1 \overset{A}{\approx} 2^{2\log n}$

Take log on both sides

$\log_2(n^2)$ $\quad$ $\log_2(2^{2\log n})$

$2 * \log_2 n$ $\quad$ $2\log n * \log_2 2$

$2\log_2 n = 2\log_2 n$

log

$2^n = 4^n$

$(2^2)^n$

2^{2n}

$(2^n)^2$

2^n $\quad$ $2^n \times 2^n$

1 $\quad$ 2^n

[MCQ]

#Q. Consider the following functions:

$f_1 = 2^n$

$f_2 = n!$

$f_3 = n^n$

$f_4 = e^n$

What is the correct increasing order of above function?

A $f_1\ f_4\ f_2\ f_3$ ✓

B $f_2\ f_1\ f_4\ f_3$ ✗

C $f_2\ f_4\ f_1\ f_3$ ✗

D $f_2\ f_1\ f_4\ f_3$ ✗

Ans: A

$f_1 \to 2^n$

$f_2 \to n!$

$f_3 \to n^n$

$f_4 \to e^n$

Imp $n! < n^n$

$n(n-1)(n-2)\cdots 1 < n \times n \times n \cdots$

$n! < n^n$

$n^n = n^n$

$n^n > n! > (e^n, 2^n)$

$e \approx 2.7$

$(2.7)^n > 2^n$

$n^n > n! > e^n > 2^n$

$f_3 > f_2 > f_4 > f_1$

$f_1 \quad f_4 \quad f_2 \quad f_3$



THANK - YOU